(Please write your Enrollment Number)

Enrollment No._____

MID-TERM EXAMINATION
(Course Name : B. Tech. ECE/ECE-AI/Reappear) (Semester : 4th)
(March, 2024) OFF LINE mode

Subject Code: BEC-202	**Subject: Linear Integrated Circuits (LIC)**
Time : 1 ½Hours	**Maximum Marks : 30**
Note: Q. 1 is compulsory.	

Q1			
	(a) Define Common Mode Rejection Ratio (CMRR). Give its significance.	**(2.5*4)** UNIT-1	
	(b) What is input bias current? How the effect of input bias current is compensated?	UNIT-1	
	(c) Define hysteresis width. What are different parameters that determine the hysteresis width?	UNIT-2	
	(d) List the different types of comparators with their i/p vs o/p waveforms.	UNIT-2	
Q2	**(Attempt any Two Parts) UNIT-1**		**(5,5)**
	(a) Explain Pole-Zero compensation technique. Compare it with Dominant-Pole compensation technique by plotting fully labelled gain vs frequency curves.		
	(b) Explain the following op-amp parameters: i. Input Offset Voltage ii. Supply Voltage Rejection Ratio iii. Output Voltage Swing iv. Slew Rate		
	(C) Give the block diagram for internal structure of op-amp. Explain each block with detailed description.		
Q3	**(Attempt any Two Parts) UNIT-2**		**(5,5)**
	(a) Explain the operation of square wave generator (Astable multivibrator) by drawing the capacitor and output voltage waveforms.		
	(b) Explain the working of precision full wave rectifier with circuit diagram and draw input & output waveforms.		
	(C) Draw the circuit of an integrator and explain its operation. Give output waveforms for the integrator if the following input signals are applied:- (i) Sine wave input (ii) Step input (iii) Square wave input		